Mower WC20-1

AGENDA: KICK-OFF MEETING

PROPOSED BY: MUHAMMAD AFFANDI BIN MANSOR

DATE: 26.12.2020

Agenda

1. Purposes & Outcomes
2. Project team
3. Timeplan
4. Overview of main components
5. Product design - Functionalities/Specifications

1. Purposes & Outcomes

- Purposes
  1. Develop a mobile robot, which can carry out the job of the conventional lawnmower.
  2. Robot can either be controlled as Fernsteuerung or run autnomously.
  3. Make Weihnachtsferien Corona 2020 as fun as possible!

- Outcomes
  1. Exposure in working with Microcontroller and electronics in a real-life/physical project.
  2. Knowledge and price estimation in order to build a larger robot or robot with other purposes.
  3. Several cutting mechanisms that can be used in this project or other projects in future.
  4. A robot that is ready and requires minimum human attention/effort to do the real task

2. Project team

Name	Role
Muhammad Affandi Bin Mansor	Project leader • Project champion • Mechanical testing and data analysis • Support mechanical and electrical tasks • Programming microcontroller
Muhamad Zainul Hakim Bin Dalim	Lead mechanical design • Mechanical design (CAD Model) • Support electrical tasks • Programming microcontroller
Muhammad Ashraf Bin Mansor	Lead electrical • Design electrical circuit • Programming microcontroller • Support mechanical tasks

3. Timeplan

Mower-WC 20-1

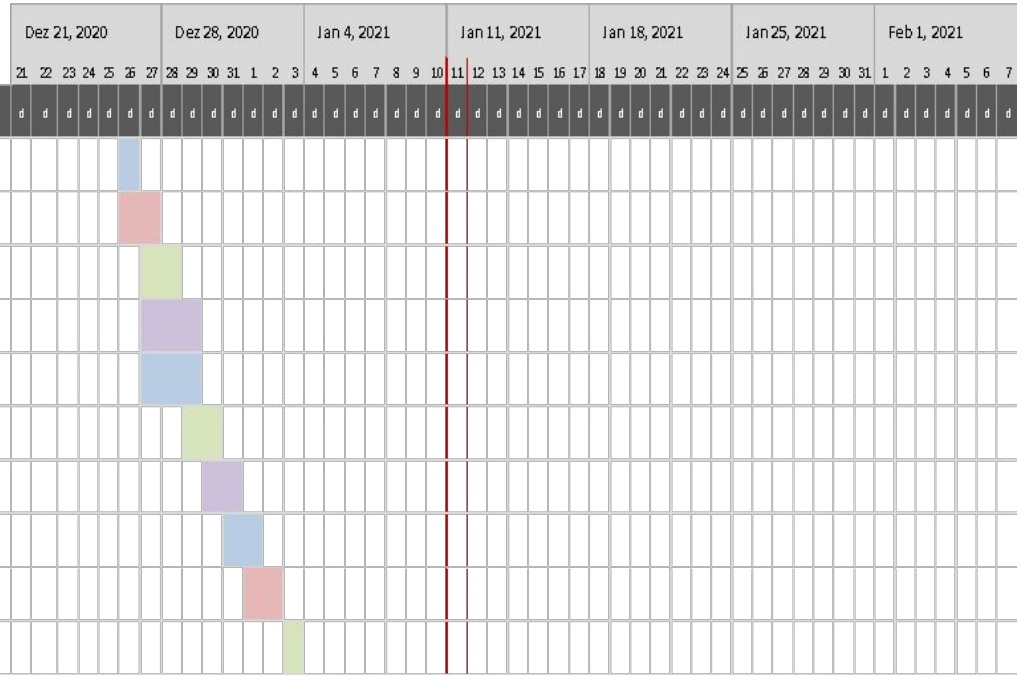
Company Name
Project Lead

Project Start: Sa, 12.26.2020

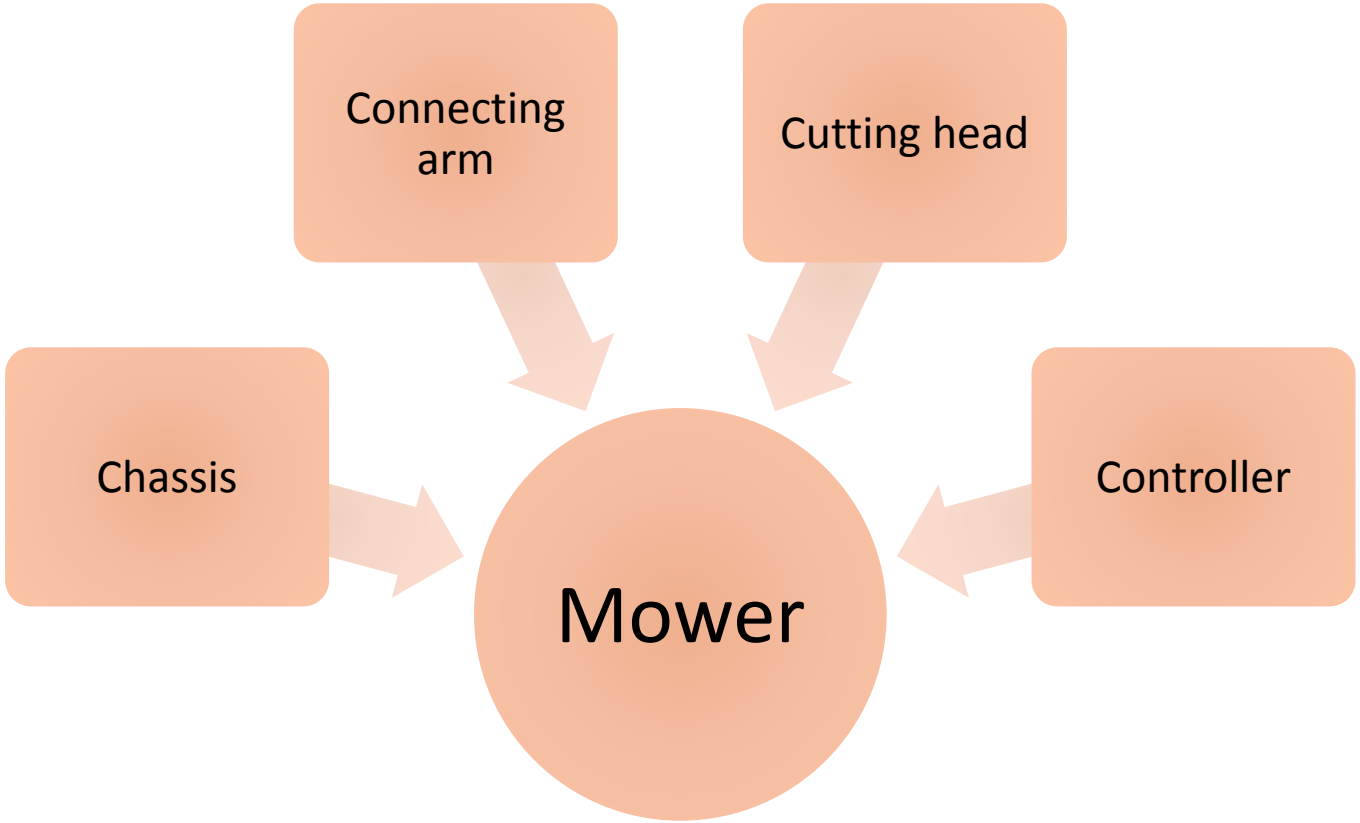
Display Week: 1

SIMPLE GANTT CHART by Vertex42.com
https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html

TASK	ASSIGNED TO	REMARKS	PROGRESS	START	END	DAYS
Project Kick-Off Meeting						
Product design review						
Change of the product design						
3D Model construction						
Designing the electrical circuit						
Microcontroller programming & testing						
Hardware construction						
Components assembling						
Product testing						
Project revision						



4. Overview of main components



5. Product design - Functionalities/Specifications

1. General
   i. Robot chassis moves in 4 directions: front, back, left, right.
   ii. Chassis is able to run autonomously or by remote control.
   iii. Cutting grass on flat and if possible, on uneven surface.

2. Chassis
   i. Housing the "brain", power pack and pivot point of the connection arm
   ii. Equipped with on/off switch
   iii. Has a wireless connection to the controller
   iv. Indicates the direction of the movement through, i.e. LED, Buzzer
   v. Sends a signal to the cutting head if the chassis wants to reverse

5. Product design - Functionalities/Specifications

3. Connecting arm
   i. Connecting the chasis and cutting head
   ii. Able to move up and down
   iii. Sleep when not in use, i.e. use driver A4988
   iv. Rotatable platform max. 90°
   v. Moves upward (i.e. 5cm from current position) when the cutting blade in operation and the chasis wants to reverse

4. Cutting head
   i. Always parallel to the earth/cutting surface
   ii. The diameter of the cutting blade can be varied
   iii. Safe enough to contain the blade if it is loosen off
   iv. Buzzer (and sensor) to indicate when the motor is stuck
   v. Equipped with caster wheels to support the turning manouver of the chasis
   vi. When in operation or turned on: Stop spinning and send the signal to the connection arm to pull upward if chasis wants to reverse

5. Product design - Functionalities/Specifications

5. Controller
   i. Has a wireless connection to the chassis/car
   ii. Can be switched on/off
   iii. Controls the movement of the car:
      a) Forward, backward, left, right
      b) Auto forward
   iv. Controls the cutting blade:
      a) Switch on/off
      b) Variable speed
   v. Controls the connection arm:
      a) Switch on/off
      b) Move up and down
   vi. Indicates when functions are on/off through, i.e. LED
   vii. Indications – Popup display, Battery status, mode of operation (autonomous), low bat warning
   viii. Emergency stop button - cut off electric supply to the cutting motor and driving motors

„The distance between you and the next Tony Stark is only an action. Lucky for you because in this project you are forced to take actions!“

Thank you for your attention!